How TiDE makes a forecast

Das et al. (2023) · supervised baseline trained on each dataset · Darts TiDEModel

1. A window of inputs

Each forecast reads a window of the last 400 days (36 months). A ? is a value not known yet.

past *future*

Target temperature

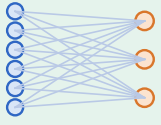
Known ahead day length

Past only NAO index

Prices go in as log differences, oil and yields as differences; all inputs are min-max scaled on the training rows.

2. Covariates to 4 numbers

A small residual network turns each step's covariates into 4 numbers, for the window and, for known-ahead ones, the horizon too.

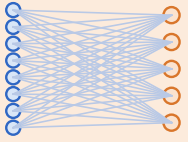


covariates *4 numbers*

This keeps the flattened input small when there are many covariates (seven past-only for temperature).

3. Dense encoder and decoder

The window and the projected covariates are flattened into one long vector. Residual MLPs encode it and decode 16 numbers per future step.



flattened input *per future step*

Temporal decoder: each step's 16 numbers plus that step's known-ahead covariates → the value

Linear shortcut: the window mapped straight to the horizon and added

4. Train once, reuse

Trained once per dataset on the rows before the first test window (daily: the last 4,096), then every window is forecast from its own origin with no retraining.

point forecast

Optuna tries a few settings on validation windows inside the training rows; the first is the default. Hidden size 128 by default (tuned 32 to 128); 10 epochs, MSE loss.

No attention and no recurrence: plain dense layers, which makes TiDE fast to train.

1 encoder and 1 decoder layer · window 400 days or 36 months · direct forecast of 90 days or 24 months · tuned: hidden size, learning rate